

Jiantong Liu

Curriculum Vitae*

Department of Mathematics
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Research Interest	Motivic Homotopy Theory, Category Theory, Algebraic K-theory, Equivariant Homotopy Theory; Brown-Peterson spectra, motivic fields, categorical algebra, tensor-triangulated geometry, power operations, synthetic spectra, applications of algebraic topology.	
Education	Ph.D. in Mathematics University of Illinois Urbana-Champaign (UIUC) Advisor: Prof. Jeremiah Heller	Aug 2023 - Present
	B.S. in Mathematics , <i>Departmental Highest Honors</i> B.S. in Cognitive Science , <i>with Specialization in Computing</i> <i>Magna cum laude</i> University of California, Los Angeles (UCLA)	Sept 2019 - Jun 2023
Publications & Preprints	Sorted by project completion date, latest first. 1. <i>Bounds in Simple Hexagonal Lattice and Classification of 11-stick Knots</i>. (With Y. Bao[†], A. Benveniste[†], M. Campisi, N. Cazet, A. Goh[†], E. Sherman[†].) <i>J. Knot Theory Ramif.</i> (2023), Vol. 32, No. 14, 2350097.	
Talks	Invited Talks ◇ <i>Bounds in Simple Hexagonal Lattice and Classification of 11-stick Knots</i>. Joint Mathematics Meeting, Jan 2023, Boston, MA. Selected Expository Talks ◇ <i>η-periodic Motivic Stable Homotopy Theory</i>. UIUC Motivic Homotopy Theory Literature Seminar, Sept 2026. ◇ <i>No Nerves of Steel</i>. UIUC Graduate Student Homotopy Theory Seminar, Mar 2026. ◇ <i>Descent Properties of Algebraic K-Theory</i>. UIUC Graduate Student Homotopy Theory Seminar, Apr 2025. ◇ <i>Categorifying Spectra</i>. UIUC Motivic Homotopy Theory Literature Seminar, Oct 2024. ◇ <i>Equivariant Cohomology</i>. UIUC Equivariant Homotopy Theory Learning Seminar, Feb 2024. ◇ <i>Triangulated Category's Christmas Wish List</i>. UIUC Graduate Algebraic Geometry and Commutative Algebra Seminar, Sept 2023. ◇ <i>Enriched Categories and Applications</i>. UCLA Directed Reading Program Colloquium, Mar 2022.	
Mentorship & Outreach	Illinois Mathematics Lab Graduate Team Leader ◇ Real Conic Bundles: an Experimental Approach Supervisor: Prof. Lena Ji ◇ Geometry, Arithmetic, and Physics of the Wythoff Array. ($\times 2$) Supervisor: Prof. Kenneth Stolarsky ◇ Stochastic Process and Differential Equation Models for Cross-Scale Evolutionary Dynamics. ($\times 2$)	Aug 2024 - Present

*Last updated October 2026.

[†]Undergraduate student at the time of article submission.

Supervisor: Prof. Daniel Cooney

UIUC Directed Reading Program

Aug 2025 - Present

Co-organizer and Mentor

- ◇ Fall 2026: Maxwell Dodson, *Commutative Algebra with a View Toward Algebraic Geometry*.
- ◇ Spring 2026: Jason Niu, *Introduction to Group Theory*.

UIUC Math Circle

Sept 2024 - Oct 2024

Instructor

UIUC Graduate Affairs Committee

Aug 2023 - May 2024

Graduate Student Member

UCLA Olga Radko Endowed Math Circle

Sept 2021 - Jun 2023

Assistant Instructor

Teaching

At UIUC

Aug 2023 - Present

Teaching Assistant

- ◇ MATH 314, *Introduction to Higher Mathematics*.
- ◇ MATH 257, *Linear Algebra with Computational Applications*. (×2)
- ◇ MATH 231E, *Calculus II*.
- ◇ MATH 227, *Linear Algebra for Data Science*.
- ◇ MATH 221, *Calculus I*.

Course Grader

- ◇ MATH 444, *Elementary Real Analysis*.
- ◇ MATH 241, *Calculus III*.

At UCLA

Sept 2020 - Sept 2022

Course Grader

- ◇ MATH 115AH, *Upper Division Linear Algebra (Honors)*. (×5)
- ◇ MATH 115A, *Upper Division Linear Algebra*. (×2)
- ◇ MATH 105, *Mathematics and Pedagogy for Teaching Secondary Mathematics*. (×3)

**Conference &
Workshop
Attendance**

Northeast Conference on Categorical Methods. Cambridge, MA.

Nov 2026

Motivic Homotopy Theory: Connections and Applications. Berkeley, CA.

Oct 2026

Representation Theory Under the Influence of Quantum Field Theory
& Motivic Homotopy Theory. Berkeley, CA.

Aug 2026

Motives in Toronto. Toronto, ON, Canada.

Jun 2026

Homotopy Theory, K-theory, and Topological Data Analysis. London, ON, Canada.

Jun 2026

A conference in honor of Rick Jardine.